Peer review
Team ROSS

- **Describe the goal of the project.**

The goal of the project is to estimate the cost of health insurance based on factors including history of cancer and/or surgeries, chronic disease status, age, and BMI. They want to do this to explore and better understand patterns in yearly insurance costs and to make predictive models that can help to estimate these premiums based on the customer's different characteristics. They also want to better inform people when they are choosing between different insurance plans.

- **Describe the data set used in the project. What are the observations in the data? What is the source of the data? How were the data originally collected?**

The data set is from an insurance company, and each observation represents a customer and their related health records/premium insurance cost. There are 986 observations and 11 columns. 10 columns dealing with customer health characteristics and one dealing with premium price, which is measured in Indian Rupees. The data was collected by the insurance company.

- **Consider the exploratory data analysis (EDA). Describe one aspect of the EDA that is effective in helping you understand the data. Provide constructive feedback on how the team might improve the EDA.**

One aspect that made sense was the correlation plot between their selected predictor variables. This allows us to see if there is any multicollinearity between their predictors. Another insightful observation was their section on "Should age be a continuous or categorical predictor?" The scatterplot between age and insurance premium (1k) shows clear clusters/buckets between different age groups. This led them to question if age should be continuous or categorical (ie, these different age groups). They further observe any possible interactions on age with a History of cancer, which reveals a clear separation between these clusters in the scatterplot.

The various EDA plots that they have are very useful. There are two ways they can improve their EDA: they can add more graphs showing the relationship between other predictors and the outcome variable (ex: bmi and price_1k) and they could continue to play with the aesthetics of their EDA graphs to make them more "pretty".

- **Describe the statistical methods, analysis approach, and discussion of model assumptions, diagnostics, model fit.**

To decide between age being a continuous or categorical predictor, they made two models with one with age being continuous and the other being categorical. They compared the R^2, AIC, and BIC of each model and settled upon age being categorical.

They also looked into if the assumption of constant variance was true. They observed non-constant variance in price with age. To test this, they made two GLS models with different sigma matrices, the first being a unique variance to each age group and the second with a variance that increases accordingly to a power function of age. From here, the model with the lower AIC and BIC was the one with a unique variance for each age group. They settled on this sigma matrix.

They then looked at different models to help choose their predictors: OLS with interaction effects, ridge regression, and LASSO. After comparing these different models, they settled on the variables chosen by LASSO due to its lower AIC and BIC.

Their final GLS model included age as a categorical variable, non-constant variance by age groups, and the predictors chosen from their LASSO regression. They then talked about the most important predictors in their GLS model.

- **Provide constructive feedback on how the team might improve their analysis. Make sure your feedback includes at least one comment on the statistical modeling aspect of the project, but also feel free to comment on aspects beyond the modeling.**

I enjoyed their project in how they used various types of models that we learned in class, and breaking down assumptions to help choose these “best” predictors for their goal. To improve their analysis, I would include short summaries of what the different models' methods are doing and why they are using them. For example, we learned in class that Ridge Regression was used for datasets where there were absurdly more columns compared to observations. However, in this dataset, that wasn’t the case. That would make sense why LASSO regression did better than Ridge. Does it make sense to use Ridge Regression as a model to compare with in this case?

- **Provide constructive feedback on the interpretations and initial conclusion. What is most effective in the presentation of the results? What additional**

detail can the team provide to make the results and conclusions easier for the reader to understand?

In the final paragraphs of their report, they included they effectively shared their results of the final GLS model and what variables were statistically significant. In hopes of making these results better to understand, I would suggest having a table with the R^2 values and what other metrics they used to decide which variables were most statistically significant. I think it would make these results more readable to the reader.

- **What aspect of this project are you most interested in and think would be interesting to highlight in the written report?**

In this project, I was most interested in their use of different models to work down to what predictors they wanted in their final GLS model. I would make sure to include more dialogue on why they chose these models and what they do in their project: Ridge Regression, Lasso, and OLS.

- **Provide constructive feedback on any issues with file and/or code organization.**

There were no big issues with the code or organization. I noticed that there were a lot of pages full of the raw output of their models. Is there any way they could highlight the information that they want to get across without having to dump out all the tables and fill the pages?

- **(Optional) Any further comments or feedback?**

It was a great project and I loved the statistical procedures they took to get to their final model.